

FIELD REFERENCE

The Photographer's Cheat-Sheet Pack

Grab-and-Go Cards for Every Shooting Situation

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What's Inside

Grab-and-Go Cards for Every Shooting Situation. Written for new camera owners who want a printable field reference.

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How to Use This Pack

This is not a book you read once and shelve. It's a deck of field cards — each one dense, self-contained, and built to live in your bag. Print them, laminate them, and pull the right card the moment you're standing in front of a scene and blanking on where to start.

Every card that follows is a standalone reference. There's a card for exposure, a card packed with ready-to-shoot recipes, and one card each for aperture, shutter, ISO, white balance, autofocus, and composition — plus a troubleshooting card for when a shot comes out wrong. Nothing depends on anything else. Grab the one you need.

HOW TO GET THE MOST FROM THIS PACK

Print each card, keep them in your camera bag, and reach for the recipe card first — it turns nine common scenes into settings you can dial in without thinking.

Three ways to use these cards



Print & laminate

Each **card starts on its own page**, sized to trim down and slip into a pocket. Laminate them so they survive rain, sweat, and the bottom of your bag.



Keep on your phone

Save the PDF to your phone. When you're out and unsure, the recipe and troubleshooting cards are two taps away.



Learn by repetition

Shoot the recipes as written, then adjust. After a few outings the numbers live in your hands and you won't need the card.



Works with any camera

Every setting here applies to DSLRs, mirrorless bodies, and most phones in Pro mode. Where a control has different names by brand, both are given (for example, "Continuous AF (AF-C / AI Servo)"). Treat every number as a *starting point*, not a law — check your shot and nudge from there.



The one habit that ties it all together

Before every shot, glance at three things: shutter (is it fast enough to be sharp?), aperture (is enough in focus?), and ISO (is it as low as the light allows?). These cards exist to make that three-second check automatic.

CARD 1

The Exposure Triangle

Three controls decide how bright your photo is — and each one also changes *how the picture looks*. That second effect is the whole game. Change one, and you trade for a side effect somewhere else.

Aperture

The hole in the lens. **Lower f-number = bigger hole = more light** and a blurrier background. Controls *depth of field*.

Shutter

How long the sensor is exposed. **Faster = less light** but freezes motion. Controls *motion blur*.

ISO

Sensor sensitivity. **Higher = brighter** but grainier. Controls *noise*.

What each control does — and what it costs

Control	Typical range	Turn it up to...	Side effect
Aperture	f/1.8 – f/16	Blur background (low f) / keep all sharp (high f)	Depth of field shrinks or grows
Shutter	30s – 1/4000s	Freeze motion (fast) / show motion (slow)	Motion blur & hand-shake risk
ISO	100 – 12800	Brighten in dark scenes	Noise / grain increases

The "stop" is the shared unit

A **stop** is a doubling or halving of light. Open one stop of aperture, and you can use a shutter twice as fast — or an ISO half as high — for the same brightness. That's how you trade between the three without changing exposure.

THE EXPOSURE TRIANGLE IN ONE LINE

Pick the look you need first — frozen or flowing, deep or dreamy — then balance the other two to keep the brightness right.



Which one to set first

Decide what matters most in the scene. Motion? Set shutter first (use S / Tv mode). Background blur or sharpness front-to-back? Set aperture first (use A / Av mode). Let the camera and Auto ISO handle the rest.

CARD 2

Settings Recipes

Nine everyday scenes, each with a starting point you can dial in fast. Set these, take a shot, check the brightness, and nudge. When light is dim, let Auto ISO climb before you slow the shutter below the numbers here.

Portrait (one person)

STARTING POINT

Aperture f/2.8

Shutter 1/200s

ISO 100-400

AF Eye / Single

Wide aperture melts the background. Focus on the near eye. Go f/4 for two people so both stay sharp.

Landscape

STARTING POINT

Aperture f/8-f/11

Shutter 1/125s

ISO 100

Focus 1/3 into scene

Narrow aperture keeps everything sharp. On a tripod, drop the shutter and keep ISO at base.

Sports / action

STARTING POINT

Aperture f/4

Shutter 1/1000s+

ISO Auto

AF Continuous + burst

Shutter Priority (S / Tv). Freeze the peak moment; let ISO rise to keep the shutter fast.

Indoor / party

STARTING POINT

Aperture f/1.8-f/2.8

Shutter 1/125s

ISO 1600-6400

WB Auto / Tungsten

Open up, embrace high ISO. Keep shutter at 1/125s so people don't smear. Find a window or lamp.

Sunset / golden hour

STARTING POINT

Aperture f/8

Shutter 1/250s

ISO 100-200

WB Cloudy / Shade

Dial exposure down 1/3–2/3 stop to keep the sky rich. Warm WB deepens the glow.

Night city

TRIPOD

Aperture **f/8**

Shutter **2-10s**

ISO **100**

Timer **2s**

On a tripod, long exposure turns traffic into light trails. Use the timer so pressing the shutter doesn't shake it.

Waterfall / silky water

TRIPOD

Aperture **f/11-f/16**

Shutter **1/2-4s**

ISO **100**

Filter **ND if bright**

Slow shutter blurs water to silk. In daylight you'll need an ND filter to get the shutter that slow.

Product / still life

STARTING POINT

Aperture **f/8**

Shutter **1/125s**

ISO **100**

Focus **Single-point**

Sharp front-to-back detail. Soft window light on one side; a white card bounces fill on the other.

Pets

STARTING POINT

Aperture **f/2.8-f/4**

Shutter **1/500s**

ISO **Auto**

AF **Continuous / Animal Eye**

Get down to their eye level. Fast shutter for the sudden zoomies; continuous AF tracks the wander.

Build your own recipe in ten seconds

Ask: *Is anything moving?* Set the shutter to freeze it. *How much do I want in focus?* Set the aperture. *Is it bright enough?* Raise ISO last. Three questions, in that order, and you've built a recipe for any scene not on this card.

CARD 3

Aperture & Depth of Field

Aperture is the f-number. It sets how much of your scene is in sharp focus — the *depth of field*. Small number, shallow focus, creamy background. Big number, deep focus, everything sharp.

f-number	Opening	Depth of field	Best for
f/1.4 – f/2	Very wide	Very shallow	Dramatic portraits, low light, background blur
f/2.8 – f/4	Wide	Shallow	Portraits, pets, subject separation
f/5.6 – f/8	Medium	Moderate–deep	Groups, street, general sharpness sweet spot
f/11 – f/16	Narrow	Deep	Landscapes, architecture, front-to-back sharp
f/22+	Very narrow	Deepest (but softer)	Rarely worth it — diffraction blurs the whole frame



Use a WIDE aperture (low f) when...

- You want a blurred, creamy background
- Light is low and you need every photon
- One subject should pop from its surroundings



Use a NARROW aperture (high f) when...

- You need everything sharp, near to far
- Shooting landscapes or group photos
- Detail across the frame matters most

THE SHARPNESS SWEET SPOT

Almost every lens is sharpest around f/5.6 to f/8. When ultimate detail matters and you don't need shallow focus, live there.



Don't chase f/22 for sharpness

Stopping down past f/11–f/16 triggers *diffraction*, which softens the entire image. To get more depth of field without going too narrow, step back or focus about a third of the way into the scene instead.



Three things change depth of field

It's not just aperture. **Getting closer** to your subject and using a **longer focal length** both shrink depth of field too. That's why a close-up portrait at 85mm f/2.8 blurs the background far more than a wide landscape shot at the same f-number.

CARD 4

Shutter Speed

Shutter speed is how long the sensor sees light. Fast freezes; slow flows. It controls both motion blur in your subject and shake blur from your own hands.

Shutter speed	Effect	Use for
1/2000s – 1/1000s	Freezes fast action	Sports, birds, cars, splashing water
1/500s – 1/250s	Freezes everyday motion	Kids, pets, walking people
1/125s – 1/60s	Sharp for still subjects	Portraits, street, handheld general
1/30s – 1/4s	Motion starts to blur	Panning, intentional blur, low light on support
1s – 30s	Long streaks & smooth water	Light trails, waterfalls, stars (tripod only)

The handholding rule

To avoid shake blur, your shutter should be **at least 1 divided by your focal length**. At 50mm, use 1/50s or faster. At 200mm, use 1/200s or faster — long lenses magnify shake as much as they magnify the subject.



Crop-sensor correction

On a crop-sensor body, multiply focal length by the crop factor first — 1.5× for most brands, 1.6× for Canon. A 50mm lens then wants about 1/80s, not 1/50s. When unsure, round up.

Shutter speeds to freeze a subject

FIELD-TESTED

Still portrait → **1/125s**

Walking → **1/250s**

Kids & pets → **1/500s**

Running / sport → **1/1000s**

Birds & cars → **1/2000s**

Motion *across* the frame needs a faster shutter than motion coming toward you. Sideways streaks the most.



Stabilization helps shake, not subjects

In-body (IBIS) or in-lens stabilization buys 2–5 stops of slower shutter for *your* shake — but it does nothing for a moving subject. Turn it off on a tripod, where it can hunt and add blur.

CARD 5

ISO & Noise

ISO brightens the photo when aperture and shutter can't gather enough light. The cost is noise — the grainy speckle that grows as ISO climbs. The rule: keep ISO as low as the scene allows, but never so low that the shot goes blurry.

Lighting	ISO to start	Notes
Bright sun	100	Base ISO — cleanest possible image
Overcast / shade	200–400	Still very clean
Indoors, good light	800–1600	Minor noise, easily usable
Dim indoors / evening	3200–6400	Visible grain, still a keeper
Very dark / night	6400–12800+	Noisy but rescues the shot

THE NOISE TRADEOFF

**A grainy sharp photo is a keeper. A clean blurry one is deleted.
When forced to choose, raise ISO and keep the shutter fast.**



Base ISO is your cleanest setting

Every sensor has a **base ISO** — usually 100 — where it produces the least noise and the most dynamic range. Use it whenever there's enough light, especially on a tripod where you don't need speed.

Auto ISO — set it and forget it

- ✓ Turn on Auto ISO so the camera raises it only as needed.
- ✓ Set a **maximum** (say 6400) so noise never runs away.
- ✓ Set a **minimum shutter speed** (a "shutter floor") so it never slows below your safe handheld speed.
- ✓ Now shoot in Aperture or Shutter Priority and let ISO float underneath.



Expose it right in camera

Noise gets worse when you brighten a dark photo later. Getting the exposure right — or slightly bright — at capture keeps noise lower than rescuing an underexposed file in editing.

CARD 6

White Balance

White balance tells the camera what "white" looks like under the current light, so colors come out true instead of orange or blue. Auto handles most scenes, but knowing the presets lets you fix — or stylize — the mood on purpose.

Preset	Kelvin (approx)	Use when...
Tungsten / Incandescent	3200K	Under warm household bulbs — removes the orange
Fluorescent	4000K	Office / store lighting — cuts the green cast
Daylight	5200K	Direct sun, midday — neutral baseline
Flash	5500K	Using a flash — matches its color
Cloudy	6000K	Overcast — adds gentle warmth
Shade	7000K	Open shade — counters the cool blue

The Kelvin scale, plainly

Low Kelvin = warm/orange light; high Kelvin = cool/blue light. To warm up a photo, set a higher Kelvin number than the actual light; to cool it down, set a lower one. That's the whole trick.

Want it warmer?

Pick **Cloudy or Shade** (or a higher Kelvin). Great for sunsets, skin tones, and cozy interiors.

Want it cooler?

Pick **Tungsten** (or a lower Kelvin). Useful for crisp blue-hour and clean, clinical looks.

Shoot RAW and stop worrying

In RAW, white balance is fully adjustable after the fact with zero quality loss. Leave it on Auto, and correct in editing later. JPEG bakes it in — so for JPEG, pick the matching preset in the moment.

Mixed light is the hard case

Warm lamps and cool window light in one frame can't both be neutral — one will cast. Pick the light on your subject and balance for that, or embrace the contrast as part of the mood. A single dominant light source is always easiest to balance.

CARD 7

Autofocus

Two decisions run your autofocus: the *mode* (does it lock once or track continuously?) and the *area* (does the camera pick the spot, or do you?). Match both to your subject and your keeper rate jumps.

Focus modes — lock vs. track

Mode	Names by brand	Use for
Single	AF-S / One-Shot	Still subjects — locks focus once, holds it
Continuous	AF-C / AI Servo	Moving subjects — refocuses as distance changes
Auto	AF-A / AI Focus	Unpredictable — switches on its own; continuous is the safer default

Area modes — who picks the spot

Area mode	What it does	Use for
Single-point	You place one small point exactly	Portraits, precise focus, off-center subjects
Zone / Group	A cluster of points covers an area	Erratic but framed action
Wide / Auto-area	Camera chooses across the frame	Fast, unpredictable subjects; snapshots
Eye / Subject detect	Finds and tracks eyes automatically	People and animals — the modern default



The focus-and-recompose trap

Locking focus, then swinging the camera to reframe can shift the sharp plane off your subject — especially at wide apertures like f/1.8. For close portraits, **move the focus point** to the subject instead of recomposing.

Back-button focus — the pro habit

By default the shutter button focuses *and* fires, and those two jobs fight each other. Back-button focus moves focusing to a rear thumb button (often "AF-ON"), leaving the shutter to only take the picture. Focus once, then fire as many frames as you like without the camera re-hunting.

- ☒ Still subject? Single AF (AF-S), single-point on the eye.
- ☒ Moving subject? Continuous AF (AF-C) + burst mode.
- ☒ People or pets? Turn on Eye / Animal detection.
- ☒ Missing focus often? Switch off full-auto area and place the point yourself.
- ☒ Ready to level up? Assign back-button focus and never look back.

CARD 8

Composition

Settings make a photo sharp and bright; composition makes it worth looking at. Six rules cover the vast majority of strong images. Learn them as defaults, then break them on purpose.

Rule of thirds

Place your subject on a third-line or where they cross, not dead center. Turn on the grid to see it.

Leading lines

Use roads, fences, and shadows to pull the eye toward your subject.

Framing

Shoot through a doorway, branch, or arch to wrap the subject and add depth.

Symmetry

Center a reflection or repeating pattern for calm, deliberate balance.

Negative space

Leave empty room around the subject to make it feel isolated and strong.

Depth

Include a near, middle, and far element so the flat frame feels three-dimensional.

Try this on your next walk

Pick one rule and shoot only that for twenty minutes — say, leading lines. Hunting for a single idea trains your eye faster than trying to remember all six at once.

Compose first, settings second. A technically perfect photo of a boring frame is still a boring photo.

Watch the edges and the background

Before you press the shutter, scan the frame's edges for distractions and the background for poles "growing" out of heads. A half-step left or right often cleans up the whole shot.

CARD 9

Troubleshooting Flowchart

Photo came out wrong? Find the symptom, apply the fix. Work top to bottom — the first matching row usually solves it.

If your photo is...	Do this
Too dark	Slow the shutter, open the aperture (lower f), or raise ISO — in that order for stills. Or nudge exposure compensation +.
Too bright	Speed up the shutter, close the aperture (higher f), or lower ISO. Or nudge exposure compensation −.
Blurry (whole frame)	Camera shake — raise shutter to the handholding rule, add support, turn on stabilization.
Blurry (moving subject)	Subject motion — use Shutter Priority, pick a freezing speed, add continuous AF + burst.
Blurry (wrong thing sharp)	Missed focus — single-point AF on the eye, or turn on Eye-AF.
Grainy	ISO too high — add light, open the aperture, or lower ISO and slow the shutter on support.
Wrong colors	White balance off — match the preset to the light (or shoot RAW and fix later).

The fix, step by step

1

Name the problem first.

Zoom in on the back of the camera and decide: brightness, sharpness, grain, or color? Don't change settings blindly.

2

Change one thing at a time.

Adjust a single control, shoot again, compare. Two changes at once hide which one worked.

3

Protect sharpness above all.

If a fix would drop your shutter below the handholding rule, raise ISO instead. Sharp-and-grainy beats clean-and-blurry.

4**Re-check after every fix.**

One change often shifts the exposure — glance at brightness again before you move on.

Before-you-shoot checklist

- ☒ ISO reset to base (100) or Auto with a sensible ceiling
- ☒ Shutter fast enough for the handholding rule and any motion
- ☒ Aperture set for the depth of field you want
- ☒ Focus mode matched to a still or moving subject
- ☒ White balance on Auto or matched to the light
- ☒ Card has space and battery is charged
- ☒ Front element clean and lens cap in your pocket

**Keep this card closest**

Of all the cards in this pack, laminate this one and the recipe card first. Between "what settings?" and "why did that come out wrong?", these two solve nearly every problem you'll hit in the field.

REMEMBER

You don't have to memorize photography — you have to carry the right card and take the next shot, and these cards make sure you always can.

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Photography guides for people who treat everything like it has a plot. · storymodeguy.com